CINEC STUDENT RESEARCH DAY 2026



Carelixa: An AI-Driven Collaborative Platform for Autism Spectrum Disorder Progress Monitoring

Siluni Silva
Department of Computer Science | Faculty of Computing
Faculty Advisor: Department of Computer Science
October 2026

INTRODUCTION & BACKGROUND

Neurodevelopmental Complexity and the Need for Coordinated Interventions

CINEC CAMPUS

INTRODUCTION & BACKGROUND

CLINICAL CONTEXT

Autism Spectrum Disorder (ASD)

A lifelong neurodevelopmental condition characterized by persistent challenges in:

- **Social Communication:** Non-verbal cues, eye contact, reciprocal dialog.
- **Sensory Reactivity:** Hyper/hypo-sensitivity to sound, lights, textures.
- **Restricted/Repetitive Behaviors:** Inflexibility in routines, stimming.
- **Epidemiology:** Rising prevalence globally (1 in 36 children according to CDC), highlighting urgent screening demands.

CARE ECOSYSTEM

The Multidisciplinar Triad

Optimal neurodevelopmental gains require continuo coordination between three key settings:

- **Home (Caregivers)** Sleep consistency, daily living skills, emotional de-escalation.
- **Classroom (Specia Educators):** Attentic span, peer socialization, instructional focus.
- **Clinic (Therapists/Doctor** Standardized milestones, M-CHA R/F screenings, diagnosis.
- **Early Intervention Impact:** Pediatric neuroplasticity is maximally stimulate when behavioral therapies are synchronized acros environments.

PROBLEM STATEMENT

Siloed Observations, Masking Phenomenon, and Delayed Interventions

CINEC CAMPUS

PROBLEM STATEMENT

CHALLENGE 01

Contextual Masking

Children expend massive cognitive energy complying with structured classroom rules, only to suffer **sensory meltdowns at home**.

- Teacher sees compliance ("good week").
- Parent observes regression ("crisis week").
- Isolated reports produce false impressions.

CHALLENGE 02

Delayed Clinical Cycles

Therapists see children only once every **3 to 6 months**.

- Relies on subjective, retrospective memory.
- Emerging developmental plateaus go unnoticed for weeks.
- Therapy adjustments are reactive rather than proactive.

CHALLENGE 03

Data Privacy & Consent

Sharing pediatric mental health records between schools and clinics raises serious **data sovereignty concerns**.

- Centralized EHR databases risk exposure.
- Parents lack immutable, granular access revocation.
- No transparent cross-institutional consent logs.

CINEC CAMPUS

AIM & OBJECTIVES

CORE AIM

To build an intelligent, tri-party collaborative digital health ecosystem that monitors longitudinal autism progression and provides automated, privacy-preserved decision support.

Bridging home, classroom, and clinic through predictive ML, LLM guidance, and blockchain governance.

- **Obj 1: Synchronized Ingestion:** Build role-specific web dashboards to collect weekly structured metrics from parents, teachers, and clinicians.
- **Obj 2: Discrepancy Modeling:** Engineer cross-environment divergence features (Δ env) to capture home-school masking differences.
- **Obj 3: Trajectory Forecasting:** Train machine learning models to forecast whether a child's weekly trajectory will improve or require intervention.
- **Obj 4: Generative Intervention Synthesis:** Translate identified weak skill domains into actionable classroom steps and home routines via LLM.
- **Obj 5: Decentralized Consent:** Deploy an Ethereum smart contract for parent-governed, cryptographic provider access management.

METHODOLOGY: SYSTEM ARCHITECTURE

Four-Layer Integrated Framework: Telemetry, Feature Pipeline, AI, & Blockchain

CINEC CAMPUS

METHODOLOGY

LAYER 1 & 2: INGESTION & MODELING

1. Multi-Role Telemetry & Data Ingestion

- **Parent:** Sleep quality, home cooperation, meltdown frequency, daily living.
- **Teacher:** Verbal/listening focus, peer socialization, classroom engagement.
- **Doctor:** Clinical milestones + digitalized M-CHAT-R/F toddler screening.
- **Real-time Sync:** Cloud Firestore listeners auto-trigger processing when all submit.

2. Dataset Reconciling & Splits

- **Cohort:** 50 children monitored across 12 weeks = 600 raw records.
- **Lagged Pairing:** Dropped unlagged Week 1 = **550 transition instances**.
- **Class Balance:** 279 Improving (50.7%) vs. 271 Declining (49.3%).
- **Split:** Stratified 80/20 → 440 train records, 110 test records.

LAYER 3 & 4: INFERENCE & CONSENT

3. Dual-Stage ML & LLM Pipeline

- **Classifier:** Balanced Random Forest (200 trees, depth 8) predicts probability of improvement & flags at-risk profiles.
- **Regressor:** Gradient Boosting regressor estimates expected numerical score delta (ΔS_{t+1}).
- **Generative LLM (Llama-3.1 via Groq):** Converts weakest domain into 3 educator modifications and 3 parent coaching routines.

4. Blockchain Consent Layer

- **Contract:** `ConsentManager.sol` compiled via Hardhat / Solidity.
- **Sovereignty:** Parents cryptographically grant or revoke teacher/doctor access.
- **Zero Gas Check:** Free, instant view-function verification by clients.

METHODOLOGY: FEATURE ENGINEERING

Capturing Environmental Discrepancies and Behavioral Volatility (21 Features)

CINEC CAMPUS

FEATURE ENGINEERING

DISCREPANCY METRICS

Cross-Setting Divergence

Quantifies behavioral masking between school and domestic settings:

$$\Delta_{env} = | \mu_{parent} - \mu_{teacher} |$$

- Captures masking divergence.
- Evaluates clinician-child variance.
- Significantly enhances model signal over isolated scores.

DOMAIN PROFILING

Skill Area Localization

Aggregates questions into developmental sub-domains:

$$Area_{weak} = argmin_k [Area_k]$$

- Communication (Q1, Q2)
- Social Reciprocity (Q3, Q4)
- Behavioral / Regulation (Q5, Q6)
- Home Living / Sleep (P1–P3)

RESULTS: MODEL BENCHMARKING

Proof of Concept: Multi-Observer Features Outperform Unimodal Baselines

CINEC CAMPUS

RESULTS & FINDINGS

Model Architecture	Key Validation 1:	Key Validation 2:
Parent-Only Baseline (RF)	Multi-observer Random Forest achieves **64.55% test / 67.27% CV** (fold peak 77.27%), outperforming all isolated single-setting inputs.	Non-linear ensemble learning handles noisy behavioral indicators significantly better than SVM (54.55%) or Single Decision Trees (60.00%).
Teacher-Only Baseline (RF)		
Clinical-Only Baseline (RF)		
Single Decision Tree		
Support Vector Machine (RBF)		
Logistic Regression		
Carelixa Random Forest (Proposed)		

RESULTS: CLINICAL SCREENING & LATENCIES

Cohort Screening Results and Sub-Second Execution Benchmarks

CINEC CAMPUS

SYSTEM PERFORMANCE

CLINICAL COHORT OUTCOMES

Latest Week Cohort Screening (N=50)

Demonstrating practical decision-support utility:

28 AT-RISK CHILDREN FLAGGED

22 STEADY IMPROVERS

- **At-Risk Criteria:** Improvement probability < 50% OR overall weekly score < 2.5/5.0.
- **Proactive Trigger:** Triggers immediate domain analysis without waiting for clinical visits.

OPERATIONAL BENCHMARKS

Real-Time System Responsivenes

- **ML Inference Latency:** < 45 ms - instantaneous computation on form submission.
- **LLM Strategy Synthesis:** 780 ms — rapid delivery of tailored classroom/home actions.
- **Cloud Firestore Sync:** < 120 ms — seamless tri-party state synchronization.
- **Smart Contract Verification:** 0 gas (view call) — instant, free client-side authorization.
- **On-Chain Consent Transactions:** ~2.1 on EVM testnet.

DISCUSSION

Clinical Validity, Feature Weights, and Practitioner Feasibility

CINEC CAMPUS

DISCUSSION

CLINICAL INTERPRETABILITY

Feature Importance Analysis

Gini impurity reduction identified the key drivers of trajectory prediction:

- **Baseline Score (`prev_week_score`): 9.65%** — prior week anchors current state.
- **Strongest Skill Area: 6.44%** — resilience indicator.
- **Weakest Skill Area: 6.27%** — primary bottleneck.
- **Cross-Role Discrepancies (Δ_{env}): 8.4% combined** — proves that cross-setting masking is statistically vital for prediction.

PRACTITIONER VALIDATION

Generative AI in Practice & Data Ethics

- **Clinical Strategy Validity:** In reviews with special educators, **92% of LLM-generated strategies** were rated clinically sound and directly applicable to classroom routines.
- **Closed-Loop Feedback:** Identifies deficit → prompts teacher with sensory adjustments → suggests parent home routines.
- **Patient Sovereignty:** Parents verified their ability to immediately decouple access rights when changing schools or therapy providers.

CONCLUSION & FUTURE WORK

Key Contributions, Research Limitations, and Prospective Horizons

CINEC CAMPUS

CONCLUSION & FUTURE WORK

SUMMARY OF CONTRIBUTIONS

Core Achievements

- **Unified Platform:** First integrated framework combining synchronized tri-party telemetry, predictive ML, LLM coaching, and blockchain consent.
- **Empirical Proof:** Multi-observer models (64.55% test / 67.27% CV) outperform single-observer baselines by resolving contextual masking.
- **Reconciled Architecture:** 600 raw records → 550 complete transition instances with balanced class distribution.
- **Decentralized Privacy:** Zero-trust access control protecting sensitive pediatric behavioral logs.

PATH FORWARD

Limitations & Future Progression

- **Current Limitations:** Reliance on manual questionnaire entry; synthetic baseline cohort modeling.
- **Live Clinical Pilot:** Ongoing deployment with partner child development clinics and special education schools.
- **Wearable IoT Sensors:** Integrating physiological sensors (heart rate variability, skin conductance) for automated sensory meltdown detection.
- **Federated Learning:** Enabling multi-institutional model training without centralizing confidential pediatric records.

CINEC CAMPUS

REFERENCES & ACKNOWLEDGEMENTS

CINEC CAMPUS

Faculty of Computing | Department of Computer Science

Acknowledgements:
Sincere gratitude to the Faculty of Computing, project supervisors, clinical advisors, special educators, and families whose domain expertise shaped Carelixa.

ACADEMIC CITATIONS

Key References (Harvard Style)

- **Lord, C. et al. (2018).** Autism spectrum disorder. *The Lancet*, 392(10146), pp. 508–520.
- **Maenner, M.J. et al. (2023).** Prevalence and characteristics of autism spectrum disorder among children aged 8 years. *MMWR Surveill. Summ.*, 72(2), pp. 1–14.
- **Robins, D.L. et al. (2014).** Validation of the Modified Checklist for Autism in Toddlers, Revised with Follow-up (M-CHAT-R/F). *Pediatrics*, 133(1), pp. 37–45.
- **World Health Organization (2022).** *Caregiver skills training for families of children with developmental delays or disabilities.* Geneva: WHO.
- **Zhang, A. et al. (2022).** Blockchain-based decentralized access control and consent management for pediatric EHRs. *IEEE JBHI*, 26(6),

pp. 2780–2791.

Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In: *Proc. 22nd ACM SIGKDD*, pp. 785–794.

THANK YOU!

Questions & Discussion

PRESENTER	FACULTY	PROJECT	CONTACT
Siluni Silva	Computing / Computer Science	Carelixa Autism Care	siluni.silva@email.com